

Computer Networks

TCS NQT - Core CS Subjects

TCS NQT Preparation Material
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Section 1: OSI & TCP/IP Models

The OSI model has 7 layers while TCP/IP has 4 layers. Understanding each layer's role and protocols is fundamental.

Q1. [Easy]

Which layer of the OSI model is responsible for end-to-end error recovery and flow control?

- A) Network Layer
- B) Data Link Layer
- C) Transport Layer
- D) Session Layer

Answer: C) Transport Layer

Explanation: The Transport Layer (Layer 4) provides end-to-end error recovery, flow control, and reliable data transfer using protocols like TCP.

Q2. [Easy]

How many layers does the TCP/IP model have?

- A) 4
- B) 5
- C) 6
- D) 7

Answer: A) 4

Explanation: TCP/IP has 4 layers: Application, Transport, Internet, and Network Access (Link).

Q3. [Easy]

Which OSI layer converts data into frames?

- A) Physical Layer
- B) Data Link Layer
- C) Network Layer
- D) Transport Layer

Answer: B) Data Link Layer

Explanation: The Data Link Layer (Layer 2) encapsulates packets into frames and handles MAC addressing.

Q4. [Medium]

Which protocol operates at the OSI Presentation Layer?

- A)** HTTP
- B)** TCP
- C)** SSL/TLS
- D)** IP

Answer: C) SSL/TLS

Explanation: The Presentation Layer (Layer 6) handles data encryption, compression, and translation. SSL/TLS operates at this layer.

Q5. [Easy]

The Session Layer (Layer 5) is responsible for:

- A)** Routing packets
- B)** Establishing, managing, and terminating sessions
- C)** Error detection in frames
- D)** Data encryption

Answer: B) Establishing, managing, and terminating sessions

Explanation: The Session Layer manages dialog control between applications, including establishing, maintaining, and terminating communication sessions.

Q6. [Easy]

Which layer of TCP/IP model corresponds to the OSI Network Layer?

- A)** Application Layer
- B)** Transport Layer
- C)** Internet Layer
- D)** Network Access Layer

Answer: C) Internet Layer

Explanation: The Internet Layer in TCP/IP maps to the Network Layer in OSI. It handles logical addressing and routing using IP.

Q7. [Easy]

Which device operates at OSI Layer 3?

- A)** Hub
- B)** Switch
- C)** Router
- D)** Repeater

Answer: C) Router

Explanation: Routers operate at Layer 3 (Network Layer) and make forwarding decisions based on IP addresses.

Q8. [Easy]

In the OSI model, which layer is closest to the end user?

- A)** Application Layer
- B)** Presentation Layer
- C)** Session Layer
- D)** Transport Layer

Answer: A) Application Layer

Explanation: The Application Layer (Layer 7) is the topmost layer and interacts directly with user applications.

Q9. [Medium]

Which PDU (Protocol Data Unit) is associated with the Transport Layer?

- A)** Bits
- B)** Frames
- C)** Packets
- D)** Segments

Answer: D) Segments

Explanation: Layer 1: Bits, Layer 2: Frames, Layer 3: Packets, Layer 4: Segments.

Q10. [Easy]

Which protocol is NOT an Application Layer protocol?

- A)** FTP
- B)** SMTP
- C)** TCP
- D)** HTTP

Answer: C) TCP

Explanation: TCP is a Transport Layer protocol. FTP, SMTP, and HTTP are all Application Layer protocols.

Q11. [Easy]

The OSI model was developed by:

- A)** IEEE
- B)** ISO
- C)** IETF
- D)** ANSI

Answer: B) ISO

Explanation: The OSI (Open Systems Interconnection) model was developed by the International Organization for Standardization (ISO).

Q12. [Medium]

Which OSI layers are combined into the Application Layer of the TCP/IP model?

- A)** Layers 5, 6, 7
- B)** Layers 4, 5, 6
- C)** Layers 6, 7
- D)** Layers 1, 2

Answer: A) Layers 5, 6, 7

Explanation: The TCP/IP Application Layer combines the Session (5), Presentation (6), and Application (7) layers of the OSI model.

Section 2: IP Addressing & Subnetting

IPv4 addressing, subnet masks, CIDR notation, and subnetting calculations are frequently tested topics.

Q13. [Easy]

How many bits are in an IPv4 address?

- A) 16
- B) 32
- C) 64
- D) 128

Answer: B) 32

Explanation: IPv4 addresses are 32 bits long (4 octets of 8 bits each). IPv6 addresses are 128 bits.

Q14. [Easy]

Which class of IP address has the default subnet mask 255.255.255.0?

- A) Class A
- B) Class B
- C) Class C
- D) Class D

Answer: C) Class C

Explanation: Class A: 255.0.0.0, Class B: 255.255.0.0, Class C: 255.255.255.0.

Q15. [Easy]

The IP address 172.16.5.10 belongs to which class?

- A) Class A
- B) Class B
- C) Class C
- D) Class D

Answer: B) Class B

Explanation: Class B ranges from 128.0.0.0 to 191.255.255.255. 172.x.x.x falls in this range.

Q16. [Medium]

What is the CIDR notation for subnet mask 255.255.240.0?

- A) /16
- B) /20
- C) /24
- D) /28

Answer: B) /20

Explanation: 255.255.240.0 = 11111111.11111111.11110000.00000000, which has 20 one-bits, so it is /20.

Q17. [Medium]

How many usable host addresses are available in a /26 subnet?

- A) 30
- B) 62
- C) 64
- D) 126

Answer: B) 62

Explanation: A /26 subnet has $2^6 = 64$ addresses. Subtract 2 (network and broadcast) = 62 usable hosts.

Q18. [Medium]

Which of the following is a private IP address range?

- A) 10.0.0.0 - 10.255.255.255
- B) 172.0.0.0 - 172.255.255.255
- C) 192.0.0.0 - 192.255.255.255
- D) 224.0.0.0 - 239.255.255.255

Answer: A) 10.0.0.0 - 10.255.255.255

Explanation: Private IP ranges: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16. Option B is incorrect as it covers more than the private range.

Q19. [Easy]

What is the broadcast address for the network 192.168.1.0/24?

- A) 192.168.1.0
- B) 192.168.1.1
- C) 192.168.1.254
- D) 192.168.1.255

Answer: D) 192.168.1.255

Explanation: For a /24 network, the broadcast address is the last address: all host bits set to 1 = 192.168.1.255.

Q20. [Medium]

Given IP 192.168.10.65/28, what is the network address?

- A) 192.168.10.0
- B) 192.168.10.48
- C) 192.168.10.64
- D) 192.168.10.65

Answer: C) 192.168.10.64

Explanation: /28 means block size = $2^4 = 16$. $65/16 = 4.06$, so network starts at $4 \times 16 = 64$. Network address is 192.168.10.64.

Q21. [Medium]

How many subnets can be created by borrowing 3 bits from the host portion?

- A) 4
- B) 6
- C) 8
- D) 16

Answer: C) 8

Explanation: Number of subnets = 2^n where n is borrowed bits. $2^3 = 8$ subnets.

Q22. [Easy]

Which IP address is a loopback address?

- A) 192.168.1.1
- B) 127.0.0.1
- C) 10.0.0.1
- D) 0.0.0.0

Answer: B) 127.0.0.1

Explanation: The entire 127.0.0.0/8 range is reserved for loopback. 127.0.0.1 is the most commonly used loopback address.

Q23. [Medium]

What is the maximum number of hosts in a Class A network?

- A) 254
- B) 65,534
- C) 16,777,214
- D) 16,777,216

Answer: C) 16,777,214

Explanation: Class A has 24 host bits. $2^{24} - 2 = 16,777,214$ usable hosts.

Q24. [Medium]

The address 169.254.x.x is used for:

- A) Multicasting
- B) APIPA (Automatic Private IP Addressing)
- C) Loopback
- D) Broadcasting

Answer: B) APIPA (Automatic Private IP Addressing)

Explanation: When DHCP fails, devices self-assign an address from 169.254.0.0/16 (APIPA range).

Q25. [Easy]

In IPv6, how many bits is an address?

- A) 32**
- B) 64**
- C) 96**
- D) 128**

Answer: D) 128

Explanation: IPv6 addresses are 128 bits long, written as 8 groups of 4 hexadecimal digits.

Q26. [Hard]

A company needs 500 host addresses. What is the minimum subnet mask?

- A) /22**
- B) /23**
- C) /24**
- D) /25**

Answer: B) /23

Explanation: /23 gives $2^9 - 2 = 510$ usable hosts. /24 gives only 254, which is insufficient.

Q27. [Medium]

What is the subnet mask for a /20 network?

- A) 255.255.240.0**
- B) 255.255.248.0**
- C) 255.255.224.0**
- D) 255.255.192.0**

Answer: A) 255.255.240.0

Explanation: /20 = 20 ones: 11111111.11111111.11110000.00000000 = 255.255.240.0.

Section 3: Transport Layer

TCP and UDP are the key transport layer protocols. Understand their differences, handshakes, and control mechanisms.

Q28. [Easy]

Which protocol provides reliable, connection-oriented communication?

- A) UDP
- B) TCP
- C) IP
- D) ICMP

Answer: B) TCP

Explanation: TCP (Transmission Control Protocol) is connection-oriented and provides reliable delivery with acknowledgments.

Q29. [Easy]

How many steps are in a TCP connection establishment (handshake)?

- A) 2
- B) 3
- C) 4
- D) 5

Answer: B) 3

Explanation: TCP uses a 3-way handshake: SYN, SYN-ACK, ACK to establish a connection.

Q30. [Medium]

TCP connection termination uses how many steps?

- A) 2
- B) 3
- C) 4
- D) 5

Answer: C) 4

Explanation: TCP uses a 4-way handshake for termination: FIN, ACK, FIN, ACK.

Q31. [Easy]

Which of the following is NOT a feature of UDP?

- A)** Connectionless
- B)** Low overhead
- C)** Guaranteed delivery
- D)** Fast transmission

Answer: C) Guaranteed delivery

Explanation: UDP does not guarantee delivery. It is connectionless, low overhead, and fast but unreliable.

Q32. [Easy]

Which port number is used by HTTP?

- A)** 21
- B)** 25
- C)** 80
- D)** 443

Answer: C) 80

Explanation: HTTP uses port 80. HTTPS uses 443, FTP uses 21, SMTP uses 25.

Q33. [Hard]

Which TCP congestion control algorithm uses additive increase and multiplicative decrease?

- A)** Slow Start
- B)** AIMD (Congestion Avoidance)
- C)** Fast Retransmit
- D)** Fast Recovery

Answer: B) AIMD (Congestion Avoidance)

Explanation: AIMD increases the congestion window by 1 MSS per RTT (additive) and halves it on congestion (multiplicative decrease).

Q34. [Medium]

The well-known port numbers range from:

- A)** 0-255
- B)** 0-1023
- C)** 0-65535
- D)** 1024-49151

Answer: B) 0-1023

Explanation: Well-known ports: 0-1023, Registered ports: 1024-49151, Dynamic/Private: 49152-65535.

Q35. [Easy]

Which protocol is preferred for real-time video streaming?

- A)** TCP
- B)** UDP
- C)** SCTP
- D)** DCCP

Answer: B) UDP

Explanation: UDP is preferred for real-time applications like video streaming because low latency is more important than reliability.

Q36. [Medium]

In TCP, what is the purpose of the window size field?

- A)** Error detection
- B)** Flow control
- C)** Routing
- D)** Encryption

Answer: B) Flow control

Explanation: The window size field in TCP header is used for flow control, telling the sender how much data the receiver can accept.

Q37. [Hard]

What happens during TCP Slow Start?

- A)** Congestion window increases linearly
- B)** Congestion window increases exponentially
- C)** Congestion window decreases
- D)** Congestion window stays constant

Answer: B) Congestion window increases exponentially

Explanation: In Slow Start, the congestion window doubles every RTT (exponential growth) until it reaches the threshold.

Section 4: Network Layer

The Network Layer handles routing, logical addressing, and path determination using protocols like RIP, OSPF, and BGP.

Q38. [Easy]

Which routing protocol uses hop count as its metric?

- A) OSPF
- B) BGP
- C) RIP
- D) EIGRP

Answer: C) RIP

Explanation: RIP (Routing Information Protocol) uses hop count with a maximum of 15 hops.

Q39. [Medium]

OSPF is based on which algorithm?

- A) Bellman-Ford
- B) Dijkstra
- C) Floyd-Warshall
- D) Prim

Answer: B) Dijkstra

Explanation: OSPF (Open Shortest Path First) uses Dijkstra's shortest path algorithm. RIP uses Bellman-Ford.

Q40. [Easy]

Which protocol is used to find the MAC address from an IP address?

- A) RARP
- B) ARP
- C) ICMP
- D) IGMP

Answer: B) ARP

Explanation: ARP (Address Resolution Protocol) maps IP addresses to MAC addresses. RARP does the reverse.

Q41. [Medium]

BGP is classified as which type of routing protocol?

- A) Interior Gateway Protocol
- B) Exterior Gateway Protocol
- C) Link State Protocol
- D) Distance Vector Protocol

Answer: B) Exterior Gateway Protocol

Explanation: BGP (Border Gateway Protocol) is an Exterior Gateway Protocol used for routing between autonomous systems on the Internet.

Q42. [Easy]

ICMP is primarily used for:

- A) Data transfer
- B) Error reporting and diagnostics
- C) Routing
- D) Address resolution

Answer: B) Error reporting and diagnostics

Explanation: ICMP (Internet Control Message Protocol) is used for error reporting (e.g., Destination Unreachable) and diagnostics (e.g., ping, traceroute).

Q43. [Easy]

Which field in the IP header prevents packets from looping forever?

- A) Checksum
- B) TTL (Time to Live)
- C) Protocol
- D) Fragmentation Offset

Answer: B) TTL (Time to Live)

Explanation: TTL is decremented by 1 at each router. When it reaches 0, the packet is discarded, preventing infinite loops.

Q44. [Medium]

What is the maximum hop count in RIP?

- A)** 10
- B)** 15
- C)** 16
- D)** 255

Answer: B) 15

Explanation: RIP has a maximum hop count of 15. A hop count of 16 means the destination is unreachable.

Q45. [Easy]

NAT stands for:

- A)** Network Address Translation
- B)** Network Access Termination
- C)** Node Address Transfer
- D)** Network Allocation Table

Answer: A) Network Address Translation

Explanation: NAT translates private IP addresses to public IP addresses, allowing multiple devices to share a single public IP.

Q46. [Easy]

Which type of routing is performed when routers exchange information dynamically?

- A)** Static routing
- B)** Default routing
- C)** Dynamic routing
- D)** Source routing

Answer: C) Dynamic routing

Explanation: Dynamic routing uses protocols (RIP, OSPF, BGP) to automatically discover and maintain routes.

Q47. [Medium]

OSPF divides a network into:

- A)** Subnets
- B)** Areas
- C)** Domains
- D)** Zones

Answer: B) Areas

Explanation: OSPF uses areas to create a hierarchical routing structure. Area 0 is the backbone area.

Section 5: Data Link Layer

The Data Link Layer handles MAC addressing, framing, error detection (CRC, Hamming), switches, and VLANs.

Q48. [Easy]

A MAC address is how many bits long?

- A) 32
- B) 48
- C) 64
- D) 128

Answer: B) 48

Explanation: MAC (Media Access Control) addresses are 48 bits (6 bytes), typically written as six pairs of hex digits.

Q49. [Medium]

Which error detection method uses polynomial division?

- A) Parity Check
- B) Checksum
- C) CRC
- D) Hamming Code

Answer: C) CRC

Explanation: CRC (Cyclic Redundancy Check) uses polynomial division to detect errors. It is widely used in Ethernet.

Q50. [Medium]

Hamming code can detect up to how many bit errors?

- A) 1
- B) 2
- C) 3
- D) 4

Answer: B) 2

Explanation: Hamming code can detect up to 2-bit errors and correct 1-bit errors.

Q51. [Easy]

A switch operates at which OSI layer?

- A) Layer 1
- B) Layer 2
- C) Layer 3
- D) Layer 4

Answer: B) Layer 2

Explanation: Switches operate at Layer 2 (Data Link Layer) and forward frames based on MAC addresses.

Q52. [Easy]

VLAN stands for:

- A) Virtual Local Area Network
- B) Very Large Area Network
- C) Virtual Link Access Network
- D) Variable Length Address Network

Answer: A) Virtual Local Area Network

Explanation: VLANs logically segment a physical network into separate broadcast domains for better management and security.

Q53. [Medium]

Which protocol is used for VLAN tagging?

- A) IEEE 802.3
- B) IEEE 802.1Q
- C) IEEE 802.11
- D) IEEE 802.1X

Answer: B) IEEE 802.1Q

Explanation: IEEE 802.1Q adds a 4-byte VLAN tag to Ethernet frames for VLAN identification.

Q54. [Hard]

What is the minimum number of redundant bits needed for Hamming code to correct single-bit errors in a 4-bit data word?

- A) 2**
- B) 3**
- C) 4**
- D) 5**

Answer: B) 3

Explanation: Using the formula $2^r \geq m + r + 1$, where $m=4$: $2^3 = 8 \geq 4+3+1 = 8$. So $r = 3$ redundant bits.

Q55. [Easy]

ARP maps:

- A) MAC address to IP address**
- B) IP address to MAC address**
- C) Domain name to IP address**
- D) Port number to IP address**

Answer: B) IP address to MAC address

Explanation: ARP (Address Resolution Protocol) resolves a known IP address to a MAC address on the local network.

Section 6: Application Layer

Application layer protocols like HTTP, DNS, DHCP, FTP, and SMTP are essential for TCS NQT.

Q56. [Easy]

DNS resolves:

- A) IP to MAC address
- B) Domain name to IP address
- C) MAC to IP address
- D) Port to IP address

Answer: B) Domain name to IP address

Explanation: DNS (Domain Name System) translates human-readable domain names (e.g., google.com) to IP addresses.

Q57. [Medium]

Which port does DNS use?

- A) 23
- B) 53
- C) 80
- D) 443

Answer: B) 53

Explanation: DNS uses port 53 for both TCP and UDP. UDP is used for queries, TCP for zone transfers.

Q58. [Easy]

DHCP is used to:

- A) Resolve domain names
- B) Automatically assign IP addresses
- C) Transfer files
- D) Send emails

Answer: B) Automatically assign IP addresses

Explanation: DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses, subnet masks, gateways, and DNS servers.

Q59. [Medium]

FTP uses which port(s)?

- A)** Port 20 only
- B)** Port 21 only
- C)** Ports 20 and 21
- D)** Port 22

Answer: C) Ports 20 and 21

Explanation: FTP uses port 21 for control (commands) and port 20 for data transfer.

Q60. [Easy]

SMTP is used for:

- A)** Receiving emails
- B)** Sending emails
- C)** Browsing the web
- D)** File transfer

Answer: B) Sending emails

Explanation: SMTP (Simple Mail Transfer Protocol) is used for sending emails. POP3/IMAP are used for receiving.

Q61. [Easy]

HTTPS uses which port by default?

- A)** 80
- B)** 143
- C)** 443
- D)** 8080

Answer: C) 443

Explanation: HTTPS (HTTP Secure) uses port 443. HTTP uses port 80.

Q62. [Easy]

Which HTTP method is used to retrieve data from a server?

- A) POST
- B) GET
- C) PUT
- D) DELETE

Answer: B) GET

Explanation: GET retrieves data, POST submits data, PUT updates data, DELETE removes data.

Q63. [Medium]

Which protocol retrieves emails from a mail server and downloads them to the client?

- A) SMTP
- B) POP3
- C) HTTP
- D) FTP

Answer: B) POP3

Explanation: POP3 (Post Office Protocol v3) downloads emails from the server to the client. IMAP keeps emails on the server.

Q64. [Medium]

The DHCP process follows which sequence?

- A) Discover, Offer, Request, Acknowledge
- B) Request, Offer, Discover, Acknowledge
- C) Offer, Discover, Request, Acknowledge
- D) Discover, Request, Offer, Acknowledge

Answer: A) Discover, Offer, Request, Acknowledge

Explanation: DHCP uses DORA: Discover (client), Offer (server), Request (client), Acknowledge (server).

Q65. [Medium]

Telnet uses which port?

- A)** 21
- B)** 22
- C)** 23
- D)** 25

Answer: C) 23

Explanation: Telnet uses port 23. SSH (secure alternative) uses port 22.

Section 7: Network Security

Encryption, firewalls, SSL/TLS, and common attack types are important for TCS NQT.

Q66. [Easy]

Which type of encryption uses a single key for both encryption and decryption?

- A) Asymmetric encryption
- B) Symmetric encryption
- C) Hashing
- D) Digital signature

Answer: B) Symmetric encryption

Explanation: Symmetric encryption (e.g., AES, DES) uses the same key for encryption and decryption. It is faster but requires secure key exchange.

Q67. [Easy]

RSA is an example of:

- A) Symmetric encryption
- B) Asymmetric encryption
- C) Hash function
- D) Firewall

Answer: B) Asymmetric encryption

Explanation: RSA uses a public-private key pair. The public key encrypts, and the private key decrypts.

Q68. [Medium]

A firewall operates at which OSI layer(s)?

- A) Only Layer 3
- B) Only Layer 7
- C) Layers 3-7 (depending on type)
- D) Only Layer 2

Answer: C) Layers 3-7 (depending on type)

Explanation: Packet-filtering firewalls work at Layer 3-4, while application firewalls work at Layer 7.

Q69. [Medium]

SSL/TLS provides:

- A) Only encryption
- B) Only authentication
- C) Encryption, authentication, and data integrity
- D) Only data integrity

Answer: C) Encryption, authentication, and data integrity

Explanation: SSL/TLS provides all three: encryption (confidentiality), authentication (certificates), and data integrity (MAC).

Q70. [Easy]

Which attack involves intercepting communication between two parties?

- A) DDoS
- B) Man-in-the-Middle
- C) Phishing
- D) SQL Injection

Answer: B) Man-in-the-Middle

Explanation: A Man-in-the-Middle (MITM) attack intercepts communication between two parties, potentially altering the data.

Q71. [Medium]

DES uses a key of how many bits?

- A) 32
- B) 56
- C) 128
- D) 256

Answer: B) 56

Explanation: DES (Data Encryption Standard) uses a 56-bit key. AES uses 128, 192, or 256-bit keys.

Q72. [Easy]

A digital certificate is issued by:

- A)** ISP
- B)** Certificate Authority (CA)
- C)** DNS Server
- D)** Firewall

Answer: B) Certificate Authority (CA)

Explanation: A CA issues digital certificates that verify the identity of websites and organizations.

Q73. [Easy]

Which protocol provides secure remote login?

- A)** Telnet
- B)** FTP
- C)** SSH
- D)** HTTP

Answer: C) SSH

Explanation: SSH (Secure Shell) provides encrypted remote login on port 22. Telnet is unencrypted.

Section 8: Numerical Problems

Bandwidth, propagation delay, throughput, and subnetting calculation problems commonly appear in TCS NQT.

Q74. [Medium]

A network has a bandwidth of 10 Mbps and a propagation delay of 20 ms. What is the bandwidth-delay product?

- A)** 100,000 bits
- B)** 200,000 bits
- C)** 2,000,000 bits
- D)** 20,000 bits

Answer: B) 200,000 bits

Explanation: Bandwidth-delay product = Bandwidth x Propagation delay = $10 \times 10^6 \times 20 \times 10^{-3} = 200,000$ bits.

Q75. [Medium]

A packet of 1000 bytes is sent over a link with bandwidth 1 Mbps. What is the transmission delay?

- A)** 1 ms
- B)** 8 ms
- C)** 10 ms
- D)** 80 ms

Answer: B) 8 ms

Explanation: Transmission delay = Packet size / Bandwidth = $(1000 \times 8 \text{ bits}) / (1 \times 10^6 \text{ bps}) = 8 \times 10^{-3} \text{ s} = 8 \text{ ms}$.

Q76. [Hard]

If a channel has a bandwidth of 3000 Hz and SNR of 31 dB, what is the maximum data rate (Shannon)?

- A)** 10,000 bps
- B)** 20,000 bps
- C)** 30,000 bps
- D)** 60,000 bps

Answer: C) 30,000 bps

Explanation: Shannon capacity = $B \times \log_2(1 + \text{SNR})$. SNR in ratio = $10^{(31/10)} \sim 1000$. $C = 3000 \times \log_2(1001) \sim 3000 \times 10 = 30,000$ bps.

Q77. [Medium]

A network 200.1.0.0/24 is divided into 4 equal subnets. How many hosts per subnet?

- A)** 30
- B)** 62
- C)** 64
- D)** 126

Answer: B) 62

Explanation: 4 subnets need 2 extra bits: /26. Hosts per subnet = $2^6 - 2 = 62$.

Q78. [Medium]

What is the propagation delay for a 2000 km link with a propagation speed of 2×10^8 m/s?

- A)** 1 ms
- B)** 10 ms
- C)** 100 ms
- D)** 1 s

Answer: B) 10 ms

Explanation: Propagation delay = Distance / Speed = $2,000,000 / (2 \times 10^8) = 0.01 \text{ s} = 10 \text{ ms}$.

Q79. [Hard]

Using Nyquist theorem, a noiseless channel with bandwidth 4000 Hz using 4 signal levels has max data rate of:

- A)** 8000 bps
- B)** 16,000 bps
- C)** 32,000 bps
- D)** 64,000 bps

Answer: B) 16,000 bps

Explanation: Nyquist rate = $2 \times B \times \log_2(L) = 2 \times 4000 \times \log_2(4) = 2 \times 4000 \times 2 = 16,000 \text{ bps}$.

Q80. [Medium]

In a sliding window protocol, if the window size is 7, how many frames can be sent before receiving an ACK?

- A)** 6
- B)** 7
- C)** 8
- D)** 14

Answer: B) 7

Explanation: The window size determines the maximum number of unacknowledged frames that can be in transit. Window size 7 means 7 frames.

Q81. [Medium]

A router receives a packet with TTL=1. What does the router do?

- A)** Forward the packet
- B)** Discard and send ICMP Time Exceeded
- C)** Store the packet
- D)** Broadcast the packet

Answer: B) Discard and send ICMP Time Exceeded

Explanation: When TTL reaches 0 after decrement, the router discards the packet and sends an ICMP Time Exceeded message to the source.